

BANKURA, IndiaWMO: **427060**Lat: **23.383N**Lon: **87.083E**Elev: **328**StdP: **14.52**Time Zone: **5.50 (IND)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.7	53.6	39.7	36.4	78.3	42.4	40.4	73.1	5.5	66.6	5.2	68.3	0.4	N/A	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	23.7	106.5	75.4	103.2	76.9	100.3	77.2	84.5	94.2	83.7	92.9	83.0	91.3	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.3	170.1	88.5	81.6	166.6	87.7	81.0	163.2	86.9	49.1	95.4	48.3	93.3	47.5	91.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.6	3.5	48.1	110.8	1.3	3.2	47.2	113.1	46.4	115.0	45.6	116.7	44.7	119.0		
(5)			45.4	85.9	1.4	1.0	44.3	86.6	43.5	87.1	42.7	87.7	41.6	88.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	66.7	73.8	82.9	89.2	88.7	88.1	84.4	84.4	84.2	81.3	75.2	68.3
	DBStd	8.32	3.48	4.62	4.51	4.88	4.23	4.23	2.32	2.20	2.53	3.23	3.38	3.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	37	23	1	0	0	0	0	0	0	0	0	0	13
	CDD50	11176	519	667	1019	1176	1199	1143	1068	1067	1025	970	756	567
	CDD65	5737	76	248	554	726	734	693	603	602	575	505	306	115
	CDH74	66263	481	2087	6329	10007	10048	9299	7102	7035	6544	4794	1941	596
	CDH80	31877	59	813	3475	6181	5911	5106	2774	2709	2482	1734	557	76
(14) Wind	WSAvg	0.9	0.7	0.6	1.0	1.6	1.5	1.3	1.0	0.9	0.8	0.5	0.5	0.6
(15) Precipitation	PrecAvg	57.0	0.8	1.0	0.7	1.9	4.6	8.1	14.6	12.2	9.2	2.9	0.7	0.3
	PrecMax	80.8	2.9	3.0	3.4	4.7	9.7	24.2	28.5	19.5	15.0	15.9	5.1	1.8
	PrecMin	34.5	0.0	0.1	0.0	0.0	1.8	0.4	6.0	2.5	3.7	0.0	0.0	0.0
	PrecStd	14.0	0.9	0.8	0.9	1.4	2.2	5.6	6.1	4.2	3.1	3.8	1.1	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.8	95.7	103.7	111.3	108.9	107.3	95.9	94.5	95.3	94.3	90.2	85.0
		MCWB	65.5	68.0	70.8	74.6	75.3	81.1	82.1	82.8	81.6	77.7	72.4	68.3
	2%	DB	81.6	91.7	100.0	107.7	104.4	103.3	93.1	92.5	93.0	91.7	87.4	82.1
		MCWB	63.2	67.3	70.7	73.4	78.2	81.8	82.6	82.6	81.9	78.7	72.1	65.9
	5%	DB	79.1	88.0	97.2	104.5	101.2	99.7	91.2	90.8	91.2	89.8	85.5	79.7
		MCWB	62.4	66.1	70.0	73.9	80.0	82.3	82.3	82.3	81.7	78.2	71.3	65.0
	10%	DB	76.7	85.0	94.7	101.0	98.4	96.4	89.3	89.2	89.5	87.9	83.4	77.5
		MCWB	61.7	65.8	69.6	74.4	80.3	82.1	81.8	82.0	81.5	77.6	69.9	64.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.2	73.5	79.8	83.7	85.1	85.4	84.5	84.5	84.1	82.4	76.3	71.4
		MCDB	77.6	84.1	88.7	97.2	96.2	95.8	91.2	91.0	91.1	89.5	84.9	78.9
	2%	WB	67.6	72.0	77.8	82.0	84.2	84.5	83.3	83.5	83.2	81.4	75.0	69.2
		MCDB	74.8	82.3	87.4	93.4	95.7	95.5	89.9	89.6	89.7	87.9	82.5	77.1
	5%	WB	65.6	70.2	76.2	80.9	83.4	83.6	82.6	82.8	82.5	80.6	74.0	67.7
		MCDB	74.1	81.5	86.4	91.4	93.9	94.2	89.0	88.6	88.8	86.7	81.0	76.2
	10%	WB	63.8	68.2	74.5	79.9	82.4	82.7	82.0	82.3	81.8	79.8	72.8	66.2
		MCDB	72.3	79.9	86.1	90.0	92.0	92.4	87.8	87.7	87.5	85.2	79.4	73.9
(35) Mean Daily Temperature Range		MDBR	22.2	23.5	24.5	23.7	19.7	15.2	9.9	9.6	10.5	14.4	18.9	20.7
	5% DB	MCDBR	24.9	26.5	27.0	27.3	23.1	20.8	13.0	12.2	13.4	16.0	19.6	22.0
		MCWBR	10.1	8.9	8.3	8.3	7.6	6.1	4.9	4.6	4.6	5.4	6.8	8.7
	5% WB	MCDBR	20.2	22.2	24.0	23.0	19.7	17.3	10.9	10.9	11.6	12.4	15.8	18.5
		MCWBR	9.8	9.0	9.2	8.4	8.5	6.4	4.7	4.7	4.9	5.3	6.3	8.0
(40) Clear-Sky Solar Irradiance	taub	0.575	0.542	0.574	0.619	0.680	0.697	0.605	0.585	0.594	0.601	0.633	0.626	
	taud	1.735	1.846	1.807	1.767	1.691	1.676	1.874	1.941	1.901	1.834	1.689	1.680	
	Ebn at Noon	214	237	237	229	214	209	229	234	229	218	197	193	
	Edh at Noon	66	63	69	73	78	79	64	60	62	63	69	68	
(44) All-Sky Solar Radiation	RadAvg	1177	1478	1788	1977	1935	1605	1365	1423	1417	1407	1262	1110	
(45)	RadStd	86	89	77	89	115	122	98	67	75	121	67	81	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	N/A	+169